

Module-2**WATER**

Water is a liquid at standard temperature and pressure, and it exists in three states: solid (ice), liquid, and gas (vapor/steam).

Global water resources:

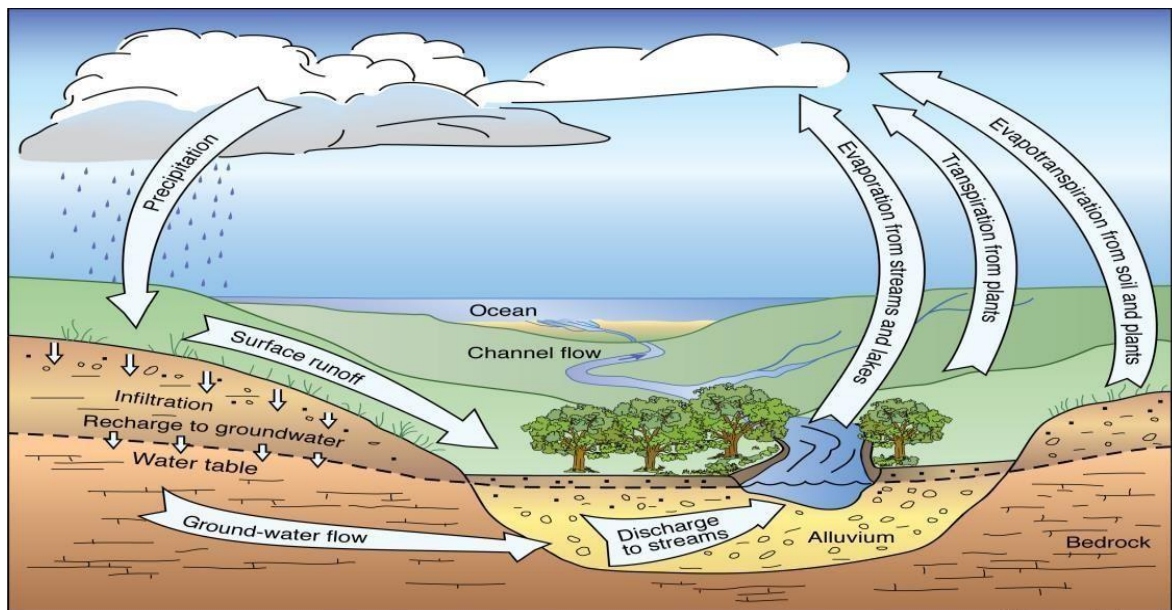
According to most estimates, about 94-97% of all water on earth is seawater held in the oceans. All freshwater resources occur in the remaining 3-6%. Around half of this is frozen in snow, glaciers and icecaps, about half is found underground as groundwater and a fraction of a percent (less than 0.1%) is surface water found in rivers, lakes, reservoirs, the atmosphere and all living things.

Water is constantly circulating between the oceans, continents and atmosphere in a system

Called the hydrologic cycle. This circulation is driven by a combination of the energy from sun, geological processes and gravity.

Water from the oceans and land is transported into the atmosphere by evaporation.

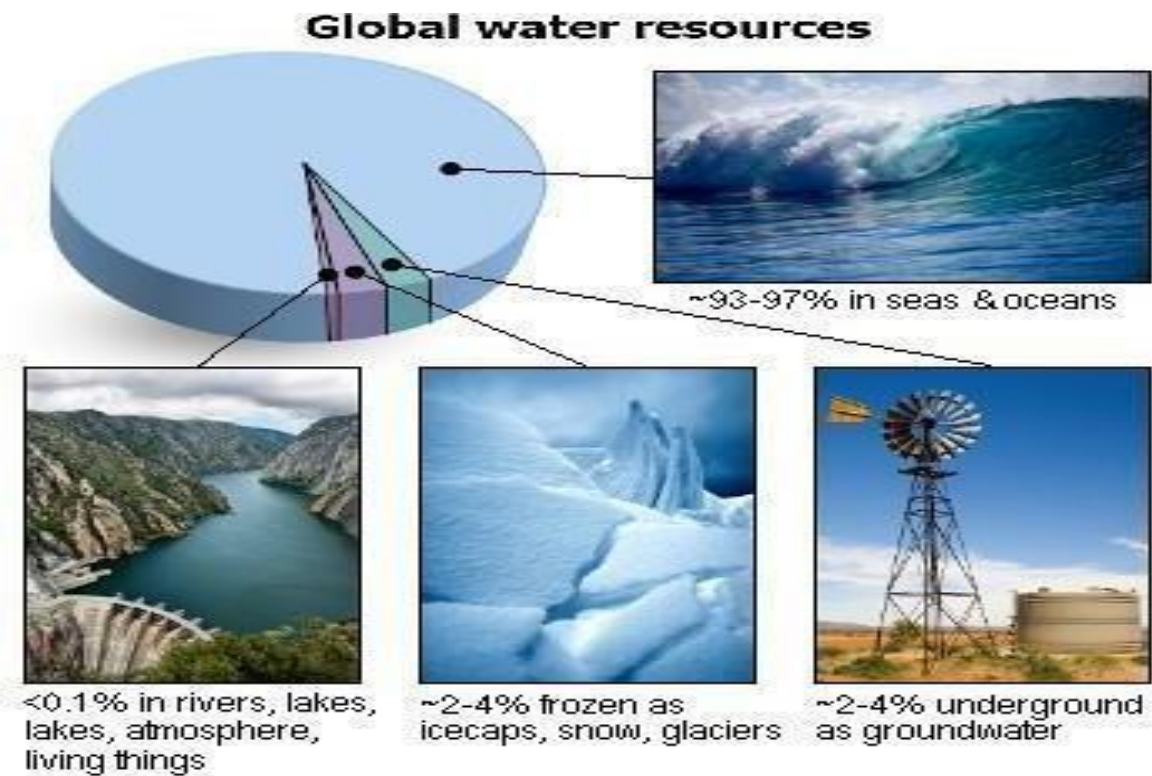
Water is also transferred to the atmosphere from vegetation by evapotranspiration.



THE HYDROLOGIC CYCLE

Water returns to the earth's surface by precipitation. On land, some of this water flows over the surface in streams and rivers into lakes, reservoirs and the oceans. The remaining infiltrates the land surface and percolates into aquifers. Some water can enter aquifers directly from the oceans. The energy of the earth's interior drives the geological processes that influence formation, influx, storage, movement and resurgence of groundwater. Groundwater can be a source of surface water on land and can flow into the oceans.

Understanding how water on land, in the oceans and in the atmosphere is connected in the hydrologic cycle is essential for the successful management of our water resources. Through these connections, factors affecting one part the cycle can influence the entire system.



Advantages of Global Water Resources

1. Essential for Life

- Supports all forms of life (plants, animals, humans).
- Crucial for drinking, hygiene, and sanitation.

2. Agricultural Use

- Irrigates crops and supports food production.
- Enables livestock farming.

3. Industrial Use

- Used in manufacturing, cooling systems, and energy production.
- Hydropower generates renewable electricity.

4. Climate Regulation

- Oceans and large water bodies regulate Earth's temperature.
- Water vapor contributes to the water cycle and precipitation.

5. Recreation and Tourism

- Supports activities like swimming, boating, and fishing.
- Attracts tourism, benefiting local economies.

Disadvantages of Global Water Resources

1. Unequal Distribution

- Some regions (e.g., deserts, parts of Africa) face water scarcity.
- Can lead to conflicts between countries or regions.

2. Pollution and Contamination

- Industrial waste, plastic, pesticides, and sewage pollute water bodies.
- Unsafe water causes diseases like cholera and dysentery.

3. Overuse and Depletion

- Over-extraction of groundwater and rivers for agriculture or industry.
- Aquifers and lakes are drying up (e.g., Aral Sea).
- 4. **Climate Change Impact**
 - Alters rainfall patterns, causes droughts and floods.
 - Melting glaciers threaten long-term water supply.
- 5. **Waterborne Diseases**
 - Lack of clean water leads to disease outbreaks in poor areas.
 - Billions still lack access to safe drinking water.

Indian Water Resources

The country houses 12 rivers that are categorized as important rivers. The overall drainage basin watered by these rivers is more than 9,76,000 sq miles or 25,28,000 km².

All the important rivers in the country have their sources in any of the following areas:

- Vindhya and Sapura Mountain range, forming a part of the heart of the country
- The Karakoram and Himalayan mountain ranges
- Sahyadri or Western Ghats Mountain range, forming a part of western India

The biggest drainage basin in the country is created by the Ganges-Brahmaputra-Meghana River network and it covers a total area of approximately 6,20,000 sq miles or 16,00,000 km². The drainage basin of the Ganga River covers a total area of around 4,20,000 sq miles or 11,00,000 km².

The source of the Ganga River is the Gangotri Glacier, which lies in the state of Uttarakhand. Subsequently, the river runs in the southeast direction, before pouring into the Bay of Bengal. The headwaters of the Ganga and Yamuna rivers lie in the western part of the Himalayan Mountain range. These two rivers meet the Ganga in the flat terrain.

The Brahmaputra River has its headwaters in the territory of Tibet. In the province of Tibet, people call it "Tsang Po" River. The river then penetrates the Indian Territory via the state of Arunachal Pradesh in the Northeast. Subsequently, the river moves to the west into Assam. In Bangladesh, the river meets the holy Ganga and the name of the river changes to the Jamuna River.

Capes in India

The major capes in India are as follows - Indira Point, the southernmost tip in India which is situated on the Great Nicobar Island, the Kanyakumari, and the southern point of territory of India., Point Claimer, and Adam's (Rama's) Bridge.

Gulfs in India

The important gulfs in India are the Gulf of Kutch, the Gulf of Cambay, and the Gulf of Munnar. The country houses straits like the Ten Degree Canal, which dissects the **Andamans** from the Nicobar Islands, the Palk Strait, which splits India from Sri Lanka, and the Eight Degree Canal, dissecting the Lakshadweep and Amindivi Islands from the Minicoy Island in the south.

The Indian Ocean and the Bay of Bengal are situated towards the south and east of the country and the Arabian Sea is located in the west. The Andaman Sea and the Lakshadweep Sea are smaller water bodies. The country houses four coral reefs and they lie in the Gulf of Munnar, the Andaman and Nicobar Islands, the Gulf of Kutch, and the Lakshadweep Islands.

Lakes in India

The major lakes in India are as follows - Vembanad Lake (Kerala), Sambhar Lake, (the biggest saltwater lagoon of the nation in Rajasthan, Loktak Lake, Kolleru Lake (Andhra Pradesh), Chilka Lake (Orissa), Dal Lake (Kashmir), and Sasthamkotta Lake (Kerala).

Water bodies that border India

Given below are the names of the water bodies that surround India from different sides:

- i) Lakshadweep Sea in the southwest
- ii) Arabian Sea in the west
- iii) Bay of Bengal in the east
- iv) Indian Ocean in the south

Water resources are natural resources of water that are potentially useful as a source of water supply. 97% of the water on the Earth is salt water and only three percent is fresh water; slightly over two thirds of this is frozen in glaciers and polar ice caps.

The remaining unfrozen fresh water is found mainly as groundwater, with only a small fraction present above ground or in the air.

Natural sources of fresh water include surface water, under river flow, groundwater and frozen water. Artificial sources of fresh water can include treated wastewater (reclaimed water) and desalinated seawater.

Uses of water: include agricultural, industrial, household, recreational and environmental activities.

Water resources are under threat from water scarcity, water pollution, water conflict and climate change. Fresh water is a renewable resource, yet the world's supply of groundwater is steadily decreasing, with depletion occurring most prominently in Asia, South America and North America, although it is still unclear how much natural renewal balances this usage, and whether ecosystems are threatened.

The framework for allocating water resources to water users (where such a framework exists) is known as water rights.

Sources of useful water-Natural sources of fresh water

Natural sources of fresh water include surface water, under river flow, groundwater and frozen water.

Surface water is water in a river, lake, wetlands. Surface water is naturally replenished by precipitation and naturally lost through discharge to the oceans, evaporation, evapotranspiration and groundwater recharge.

Although the only natural input to any surface water system is precipitation within its watershed, the total quantity of water in that system at any given time is also dependent on many other factors. These factors include storage capacity in lakes, wetlands and artificial reservoirs

- The permeability of the soil beneath these storage bodies.
- The runoff characteristics of the land in the watershed.
- The timing of the precipitation and local evaporation rates. All of these factors also affect the proportions of water loss.

Groundwater is fresh water located in the subsurface pore space of soil and rocks. It is also water that is flowing within aquifers below the water table. Sometimes it is useful to make a distinction between groundwater that is closely associated with surface water and deep groundwater in an aquifer (sometimes called "fossil water").

Groundwater can be thought of in the same terms as surface water: inputs, outputs and storage. The critical difference is that due to its slow rate of turnover, groundwater storage is generally much larger (in volume) compared to inputs than it is for surface water. This difference makes it easy for humans to use groundwater unsustainable for a long time without severe consequences. Nevertheless, over the long term the average rate of seepage above a groundwater source is the upper bound for average consumption of water from that source.

The natural input to groundwater is seepage from surface water. The natural outputs from groundwater are springs and seepage to the oceans. If the surface water source is also subject to substantial evaporation, a groundwater source may become saline. This situation can occur naturally under endorheic bodies of water, or artificially under irrigated farmland. In coastal areas, human use of a groundwater source may cause the direction of seepage to ocean to reverse which can also cause soil salinization. Humans can also cause groundwater to be "lost" (i.e. become unusable) through pollution. Humans can increase the input to a groundwater source by building reservoirs or detention ponds.

Artificial sources of usable water

Artificial sources of fresh water can include treated wastewater (reclaimed water) and desalinated seawater. However, economic and environmental side effects of these technologies must also be taken into consideration.

Water reclamation (also called wastewater reuse, water reuse or water recycling) is the process of converting municipal wastewater (sewage) or industrial wastewater into water that can be reused for a variety of purposes.

Types of reuse include: urban reuse, agricultural reuse (irrigation), environmental reuse, industrial reuse, planned potable reuse, de facto wastewater reuse (unplanned potable reuse). For example, reuse may include irrigation of gardens and agricultural fields or replenishing surface water and groundwater (i.e., groundwater recharge).

Reused water may also be directed toward fulfilling certain needs in residences (e.g. toilet flushing), businesses, and industry, and could even be treated to reach drinking water standards. Treated municipal wastewater reuse for irrigation is a long-established practice, especially in arid countries. Reusing wastewater as part of sustainable water management allows water to remain as an alternative water source for human activities. This can reduce scarcity and alleviate pressures on groundwater and other natural water bodies.

Need for Planning and Management of water resources systems are essential due to following factors:

- (1) Severity of the adverse consequences of droughts, floods and excessive pollution. These can lead to
 - a) Too little water due to growing urbanization, additional water requirements, in stream flow requirements etc. Measures should be taken to reduce the demand during scarcity times.
 - b) Too much water due to increased flood frequencies and also increase in water requirements due to increased economic development on river floodplains.
 - c) Polluted water due to both industrial and household discharges.
- (2) Degradation of aquatic and riparian systems due to river training and reclamation of floodplains for urban and industrial development, poor water quality due to discharges of pesticides, fertilizers and wastewater effluents etc.
- (3) While port development requires deeper rivers, narrowing the river for shipping purposes will increase the flood level.
- (4) River bank erosion and degradation of river bed upstream of the reservoirs may increase the flooding risks.
- (5) Sediment accumulation in the reservoir due to poor water quality. Considering all these factors, the identification and evaluation of alternative measures that may increase the quantitative and qualitative system performance is the primary goal of planning and management policies.

Resource System Planning:

Resource System Planning is the coordinated and long-term planning of resource availability, use, and development to ensure sustainable and reliable access to essential resources (such as water, energy, land, minerals, and human capital).

Components of Resource System Planning

Component	Description
Assessment of Resources	Identifying available natural and human resources (e.g., water sources, energy potential).
Demand Forecasting	Estimating future needs based on population, economy, climate, etc.
Sustainability Goals	Setting environmental, social, and economic targets (e.g. net-zero emissions, water neutrality).
Policy and Governance	Developing regulations, policies, and institutions for resource management.
Technology Integration	Using smart systems, AI, or remote sensing to improve efficiency and monitoring.
Risk Management	Preparing for uncertainties like droughts, resource conflicts, or market volatility.

Water Resource System Planning

Goals:

- Ensure reliable water supply for domestic, agricultural, industrial, and environmental needs.
- Balance surface water, groundwater, rainfall, and reuse/recycling.
- Avoid over extraction and pollution.

Steps:

1. Inventory of water sources
2. Demand analysis (current and projected)
3. Scenario modeling (e.g., under climate change)
4. Infrastructure planning – dams, canals, treatment plants
5. Policy planning – pricing, regulation, rights
6. Monitoring and evaluation

Benefits of Resource System Planning

- Prevents resource depletion and conflicts
- Ensures economic efficiency
- Enhances climate resilience
- Supports long-term sustainability
- Promotes intergenerational equity

Challenges in Resource Planning

- Data limitations
- Political and institutional barriers
- Complex interdependencies (e.g., water-energy-food nexus)
- Climate change and uncertainty
- High initial investment needs

Water Sectors in India

List of uses of water

Water is stored in various parts of the world but not evenly distributed all over the earth. It is said to be a universal solvent. Various sources of water are – sea, lake, rain, well, stream, borehole and pond. It is used for washing, drinking, generating electricity etc. Below are the different uses of water in various fields:

- Domestic uses of water
- Water uses for agriculture
- Industrial uses of water

Domestic uses of water:

15 % of water is consumed for domestic purpose. Water is used for drinking, bathing, cooking food and washing dishes, clothes, fruits, vegetables and brushing teeth.

Water use for agriculture:

Agriculture is the largest consumer of water. About 70% of water is used for irrigation. Water is necessary for gardening, farming and fisheries. Plants require water to grow. During the process of photosynthesis, they consume water. To yield crops, fruits, flowers, vegetables they need sufficient water, manure, sunlight and oxygen.

Industrial uses of water:

It is either used in creating or to cool the equipment used for creating the product. Industrial water is used for washing, cooling, processing, transporting, diluting or fabricating of a product. The maximum amount of water is used in the production of chemical, paper and food.

Other uses are – it is used in transportation, manufacturing, hydroelectric power, removal of body wastes, tourism and recreation.

Water deficits and Water Surplus

Water deficits occur when water demand exceeds supply. The disparity between human activity and water resources has resulted in increasing incidences of local and regional water crises. Collectively, this has been defined as the global water crisis.

Water Surplus precipitation exceeds evapotranspiration and the excess is not being used by plants, means there is excess water available to the system. This occurs when there is a reduction of water available within the system. This occurs when evapotranspiration exceeds precipitation.

Equitable distribution of water

Drinking water is a basic requirement of the human being. The water scarcity in urban areas of developing countries is a major concern. In future population in urban areas will face acute water problems. The daily supply of

water in developing countries is very low compared to the industrial world.

Right to water or equitable distribution needs to be seen as every community having a proportional right to water as part of a collective right to assured livelihoods. Water use beyond fulfillment of livelihood needs, does not form part of this right and moreover cannot be at the cost of other's livelihoods.

It is challenging to provide potable water in sufficient quantity, at adequate pressure and at an acceptable quality at a consumer's tap. However, with the incorporation of IoT technologies of sensing and communications, along with advanced models of hydraulics, algorithms for controls, optimization and scheduling, we believe that equity in the water distribution can be addressed.

Inter basin water transfer

The interlinking of rivers is a large-scale civil engineering project that aims to effectively manage water resources in India. The government aims to achieve this by building a network of reservoirs and canals through a **National River Linking Project (NRLP)**.

Inter basin water transfer or trans basin diversion are (often hyphenated) terms used to describe man-made conveyance schemes which move water from one river basin where it is available, to another basin where water is less available or could be utilized better for human development. The purpose of such water resource engineering schemes can be to alleviate water shortages in the receiving basin, to generate electricity or both.

Need for Inter Basin Water Transfers (IBWT)

Inter Basin Water Transfers is necessarily required to overcome the water scarcity situations in the regions/basins. These are needed to enhance water utility and reduce water wastage of water surplus areas in the following manner:

- Large variation in rainfall and available water resources in space and time
- Diversion of water from water surplus basins to water deficit basins/regions
- Use of the surplus water which is otherwise flowing into the sea unutilized
- To mitigate likely adverse impact of climate change, short term and long term

Interlinking of Rivers

Interlinking of Rivers refers to the artificial connection of rivers by man-made canals to transfer water from areas with excess water (flood-prone) to areas with water scarcity (drought-prone).

Objectives of Interlinking of Rivers

- **Mitigate** droughts and floods
- **Provide water for** irrigation and drinking
- **Improve** groundwater recharge
- **Generate** hydropower
- **Enhance** inland navigation
- **Promote** regional equity in water availability

Types of River Interlinking Projects

1. Himalayan Rivers Component

- Involves rivers like Ganga, Brahmaputra and Yamuna.
- Goal: Transfer water to north-western India and central areas.

2. Peninsular Rivers Component

- Involves rivers like Godavari, Krishna, Mahanadi and Cauvery.
- Goal: Connect east-flowing rivers in southern India.

Advantages of Interlinking of Rivers

Advantage	Description
1. Water Redistribution	Moves water from surplus to deficit areas.
2. Drought and Flood Control	Reduces floods in one area and droughts in another.
3. Agricultural Benefits	Enhances irrigation, increases cropping area and productivity.
4. Drinking Water Supply	Provides access to water in arid regions.
5. Hydropower Generation	Some projects include dams for clean energy.
6. Inland Navigation	Boosts transport and reduces traffic pressure on roads.

Disadvantages of Interlinking of Rivers

Disadvantage	Description
1. High Cost	Massive financial investment and maintenance.
2. Environmental Impact	Deforestation, loss of biodiversity, altered river ecosystems.
3. Displacement	Large-scale displacement of communities due to dam and canal construction.
4. Inter-State Conflicts	Disputes over water sharing between states or countries.
5. Climate Uncertainty	Changing rainfall patterns may affect source rivers.
6. Risk of Over engineering	Natural flow of rivers may be disrupted, harming ecology.

Example: Interlinking Projects in India

Project	Rivers Involved	Status
Ken-Betwa Link	Ken River (M.P.) → Betwa River (U.P.)	Approved, under implementation
Godavari-Krishna-Cauvery Link	Multi-phase transfer across southern India	Planned
Par-Tapi-Narmada Link	Rivers in Gujarat and Maharashtra	Facing protests

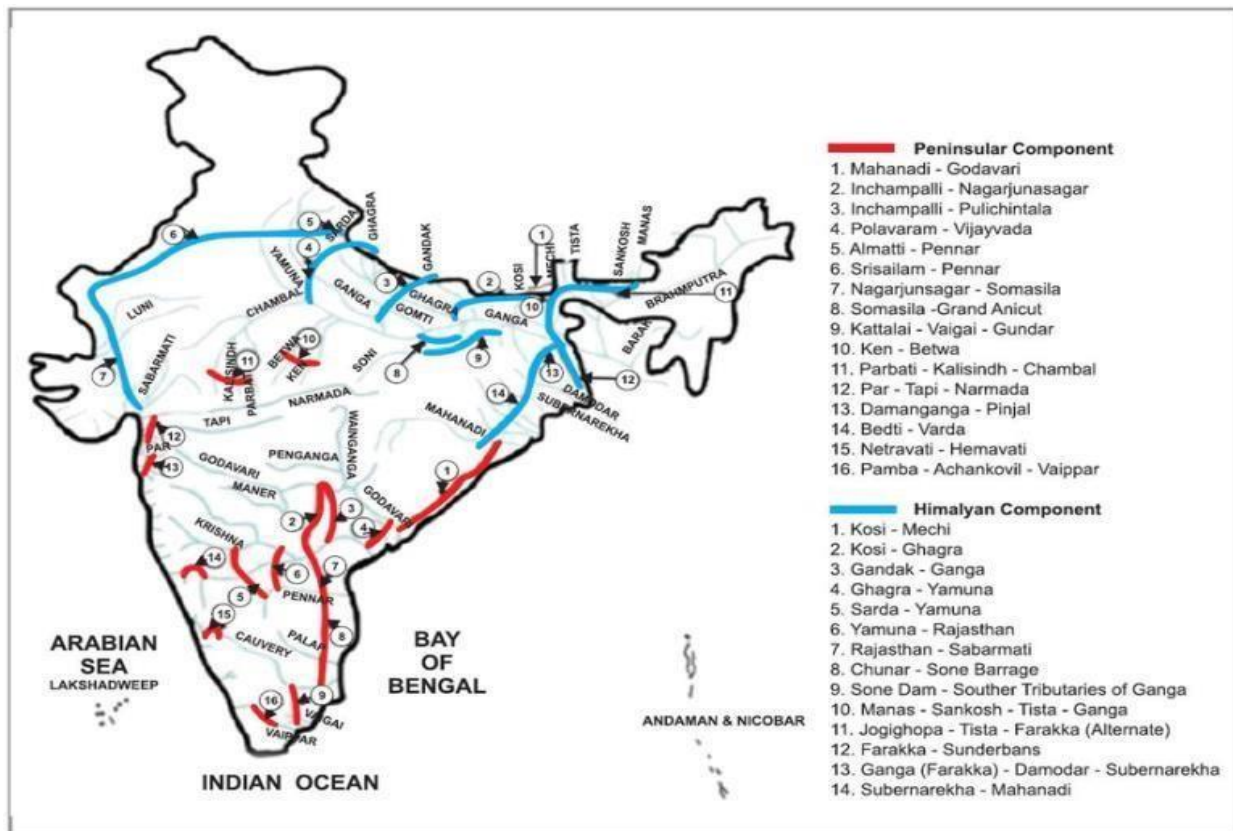
Peninsular Component of River Interlinking

(Part of India's National River Linking Project - NRLP)

The **Peninsular Component** refers to the **southern part** of the Indian subcontinent's river interlinking plan. It aims to connect major rivers in **South and Central India** to address water shortages, improve irrigation, and manage floods/droughts more effectively.

Objective:

To transfer water from water-surplus rivers (like Godavari, Mahanadi) to water-deficient basins (like Krishna, Cauvery, and Vaigai), ensuring equitable distribution of water in southern India.



Key Features:

- Covers the **southern Peninsular region of India**
- Involves **east-flowing rivers**
- Designed mostly for **irrigation, drinking water, and drought relief**
- Includes **intra-basin and inter-basin** transfers

Major Proposed Links in the Peninsular Component:

No.	Link Project Name	Rivers Involved	Purpose
1	Mahanadi-Godavari Link	Transfer from Mahanadi to Godavari	Reduce floods in Odisha, provide water to dry areas
2	Godavari-Krishna Link	Polavaram (Godavari) → Krishna River	Improve irrigation in Telangana & Andhra
3	Krishna-Pennar Link	Krishna River → Pennar Basin	Water supply to drought-prone Rayalaseema
4	Pennar-Cauvery Link	Pennar River → Cauvery Basin	Water for Tamil Nadu, reduce Cauvery disputes
5	Cauvery-Vaigai-Gundar Link	Cauvery → Vaigai → Gundar	Support irrigation and water supply in Tamil Nadu
6	Ken-Betwa Link	Ken River (M.P.) → Betwa River (U.P.)	Only link connecting central India; benefits Bundelkhand
7	Par-Tapi-Narmada Link	Par & Tapi Rivers → Narmada	Water supply to Gujarat & Maharashtra; faces protests

Advantages of the Peninsular Component

- **Increases irrigation** in drought-prone regions
- **Improves drinking water availability** in rural and urban areas
- Helps in **flood control and drought mitigation**
- Potential for **hydropower generation** in some projects
- Boosts **inland navigation and regional development**

Challenges

- **Very high cost** and need for large-scale infrastructure
- **Environmental damage**: forest loss, biodiversity impact
- **Displacement** of people and villages
- **Legal and political issues**: interstate water disputes
- **Climate change**: affects rainfall patterns and river flows

Ground water

Ground water is water that accumulates underground. It can exist in spaces between loose particles of dirt and rock, or in cracks and crevices in rocks. Different types of rocks and dirt can contain different amounts of water.

Recharge of ground water

Groundwater recharge is the process where surface water or precipitations infiltrates the soil and percolates downwards to the water table.

Groundwater Recharge is the primary method through which water enters underground resources like the aquifer.

TYPES OF GROUNDWATER RECHARGE

Groundwater recharge occurs in two forms:

- a. Natural Groundwater recharge
- b. Artificial Groundwater recharge

1. Natural Groundwater recharge

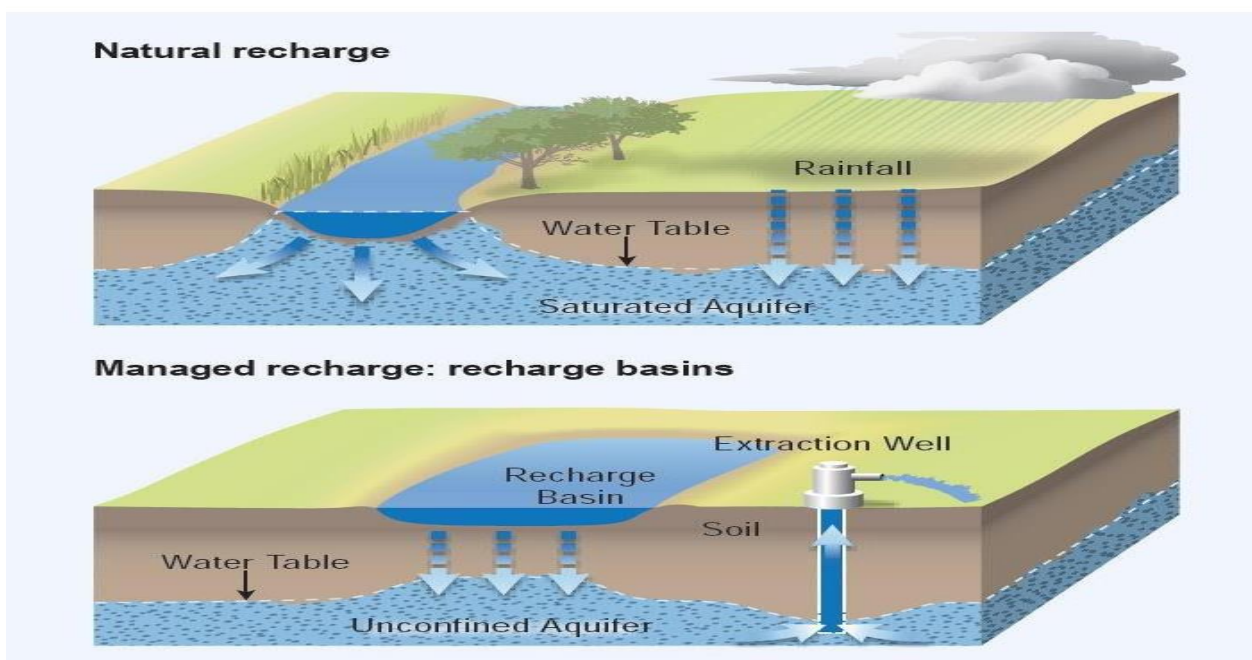
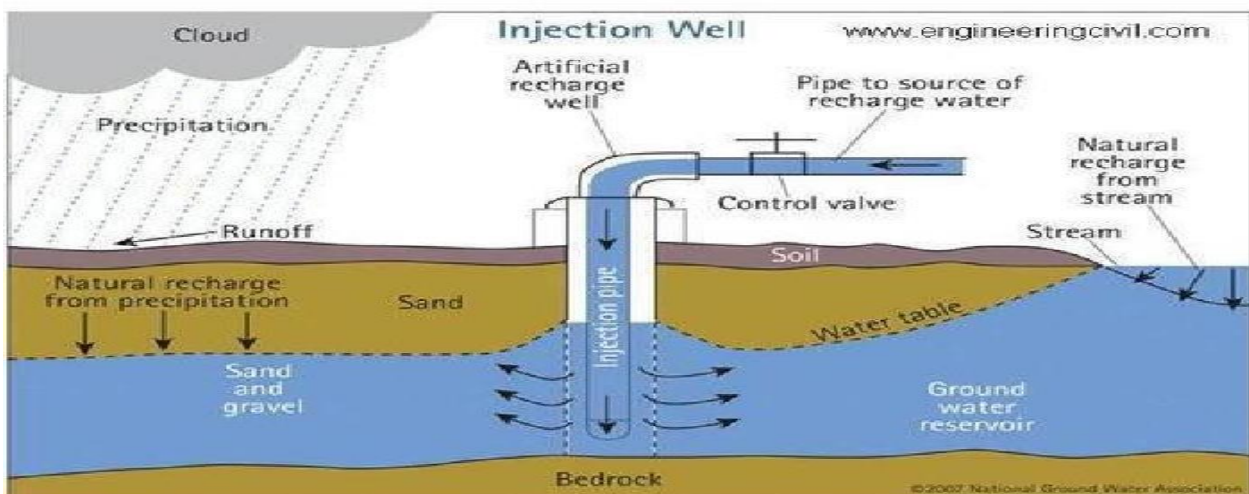
Precipitation ie. rainfall and snowmelt.

To a smaller extent by surface water like rivers and lakes.

- The water is able to move underground through the rock and soil due to connected pore spaces. This downward movement of water through different soil layers is called percolation.
- Some types of soils allow more water to infiltrate than others depending on the soil's permeability.
- During natural recharge, water is first pulled into the zone of aeration; where a mixture of water and air fills the pore space. Then the water further travels downwards to the zone of saturation – where the pore spaces are completely filled by water.
- The upper boundary of the zone of saturation is known as the water table.
- Aquifers are the underground layers of rocks that hold the groundwater and they are found in the saturation zone.

2. Artificial Groundwater recharge

- In areas where groundwater is utilized faster than its natural replenishing rate, man-made recharge method becomes a necessary option for balancing the water levels.
- Artificial recharge is the process of increasing the amount of water that enters an aquifer through planned, human-controlled means.
- Groundwater can be artificially recharged by redirecting water across the land surface through canals, infiltration basins, or ponds; adding irrigation furrows or sprinkler systems or simply injecting water directly into the subsurface through injection wells.



Factors Affecting Groundwater Recharge

Groundwater recharge depends on several factors such as:

1. Porosity and Permeability of the soil.
2. Infiltration capacity.
3. Precipitation rate.
4. Climate changes.
5. Type of vegetative cover.

Importance Of Groundwater Recharge

- Groundwater is the largest source of freshwater for mankind and approximately 30% of the freshwater on Earth is groundwater. In some rural areas, groundwater accounts for 100% of the resident's drinking water.
- It is also an integral part of the water cycle.
- The importance of groundwater recharge cannot be over-emphasized.
- For sustainable groundwater management, groundwater recharge is required to balance environmental water levels and also maintain flow into rivers, lakes and wetlands.
- Shortage of groundwater in an area will lead to depletion of aquifers and dryness of wells. Its adverse effects will be felt both on the inhabitants and the environment.

Conjunctive use

In General terms, conjunctive use implies the planned and coordinated management of surface and groundwater, so as to maximize the efficient use of total water resources.

Because of the interrelationship existing between surface and subsurface water, it is possible to store during critical periods the surplus of one to tide over the deficit of the other.

Thus groundwater may be used to supplement surface water supplies, to cope with peak demands for municipal and irrigation purposes, or to meet deficits in years of low rainfall.

On the other hand, surplus surface water may be used in overdraft areas to increase the groundwater storage by artificial recharge.

Moreover, surface water, groundwater or both, depending on the surplus available, can be moved from water-plentiful to water-deficit areas through canals and other distribution systems.

The beneficial effects of conjunctive water use in canal commands can be summarized as follows:

- Use of groundwater helps cope with peak demands for irrigation and hence reduce size of canals and consequently construction costs
- Supplemental supplies from groundwater bodies ensure proper irrigation scheduling, even if rainfall fails or is delayed.

- Groundwater withdrawals lower the water table thus reducing the risk of water-logging, soil salinization and consequent wastage of water for leaching the soils
- Surface and subsurface outflows are minimized, causing reduction in peak runoff
- When conjunctive use is integrated with artificial recharge the need for lining canals is reduced, as seepage from canals replenishes groundwater
- Conjunctive use allows the utilization of saline or brackish ground – or surface – water resources, either by mixing them with freshwater, or by using alternate water resources for irrigation.

Contamination of ground water

Groundwater pollution occurs as a result of the release of pollutants into the ground to natural underground water reservoirs known as aquifers. Once the pollutants released to find their way into groundwater, they cause contamination. It is a type of water pollution that is mainly caused by the release of substances either intentionally or accidentally through anthropogenic activities or natural causes.

The pollutants usually move within aquifers depending on biological, physical, and chemical properties. Processes such as diffusion, dispersion, adsorption, and the speed of moving water often facilitate the movement. But in general, the movement of the contaminants within an aquifer is usually slow and as such, their concentration tends to be high and in a form called a plume.

As the plume spreads it might connect with springs and ground wells making them unsafe for human consumption. Hence, this article discusses the causes, effects, and various solutions to underground water pollution.

Causes of Groundwater Pollution

1. Natural Sources

Naturally occurring substances found in the soils and rocks can be dissolved in water causing contamination. These substances are sulphates, iron, radionuclides, fluorides, manganese, chlorides, and arsenic. Others such as the decaying materials in the soil may seep in underground water and move with it as particles.

Reports by WHO indicate that the most common pollutants are fluoride and arsenic. The natural cause of pollution can be tested using the Groundwater Assessment Platform (GAP). GAP estimates contamination levels using environmental, geological, and topographical data.

2. Septic Systems

Across the world, septic systems are the main cause of pollution of underground water. The pollutants are outflow from privies, septic tanks, and the cesspools. 25% of households in the USA, for instance, heavily depend on the septic systems to dispose of their waste. This huge number of users relying on the system makes it one of the main pollutants.

Additionally, improperly designed and leaking septic systems release contaminants such as nitrates, oils, bacteria, chemicals, detergents, and viruses into underground water.

3. Hazardous Waste Disposal

Hazardous wastes such as photographic chemicals, motor oil, cooking oil, paint thinners, medicines, swimming pool chemicals, paints, and garden chemicals should not be disposed into septic tanks or directly into the environment as they cause serious contamination. These chemicals should be disposed of with the help of a licensed hazardous waste handler.

4. Petroleum Products

Petroleum storage tanks are either located underground or above ground. Also, the transportation of petroleum products is mainly done underground using the pipeline. Leakages from these substances can lead to contamination of water.

In the USA it is estimated that 16,000 chemical spills each year are from trucks, storage containers, and train spillages especially when transferring oil. The chemicals spilled become diluted with water and seep into the ground and may cause groundwater contamination.

5. Solid Waste

Palmer Developmental Group estimated that in developing countries approximately 0.3 to 0.6 kg/person/day of waste is released into the ground. On the other hand, in developed countries, 0.7 to 1.8 kg/person/day is released. The chemicals from these substances are leached into the groundwater through precipitation and surface runoff.

The wastes can also be collected and taken to landfills. If the landfills lack clay liner and leachate the chemicals from the wastes will leach and pose a threat to the groundwater.

6. Surface impoundments

These are shallow lagoons used to store liquid wastes. The USA, for example, has over 180,000 surface impoundments which can pose a threat to groundwater. Therefore, the impoundments are required to have clay liners or leachates to prevent leaching. In some cases, the leachates may be defective and leakages may occur leading to contamination of water.

7. Agricultural Chemicals

Millions of tons of agricultural chemicals such as fertilizers and pesticides are used worldwide to increase crop production. Other institutions such as the golf courses also use these chemicals.

Excessive use of these chemicals can lead to contamination of groundwater. Chemicals such as pesticides are known to remain in the ground for years and when diluted with the rainwater they seep deeper into the groundwater.

8. Injection wells

They have various uses ranging from collection or stormwater to disposal of industrial and commercial effluents. When not properly regulated, hazardous chemicals can be disposed of from injection wells. For this reason, if not properly located, regulated, and designed; they can cause contamination of groundwater.

9. Other causes

Other causes of ground pollution are abandoned wells which can act as a pathway for contaminants to reach the aquifers. Also, poorly constructed wells that may lack proper casing and covers may cause groundwater contamination that is pollutants find their way into such wells.

Another cause of pollution is mining activities where through precipitation the soluble minerals can be leached from the sites to the groundwater.





Effects of Groundwater Pollution

1. Health Issues

Contaminated groundwater has detrimental effects on health. In areas where septic tanks installation is not set up correctly, the human waste may contaminate the water source. The waste may contain hepatitis causing bacteria that may lead to irreversible damage to the liver.

Also, it may cause dysentery which leads to severe diarrhea, dehydration, and in some cases death. Additional health problems include poisoning that may be a result of the use of excessive pesticides and fertilizers or natural chemicals. The chemicals leach into water sources and poison them. Drinking water from such a source may lead to serious health effects.

2. Affects economic growth

Contamination of groundwater sources renders the area incapable of sustaining plant, human, and animal life. The population in the area reduces and the land value depreciates. Another effect is that it leads to less stability in industries relying on groundwater to produce their goods.

Therefore, the industries in affected areas will have to outsource for water from other regions which may turn out to be expensive. In addition, they may be forced to close down due to the poor quality of water.

3. Can lead to damaging impacts on the environment such as aquatic systems and the overall ecosystem

Groundwater pollution can lead to devastating environmental changes. One such alteration is the loss of certain nutrients that are essential for self-sustenance of the ecosystem. Also, when the pollutants mix with water bodies, alteration of the aquatic ecosystem may also occur. Aquatic animals such as fishes may die off quickly as a result of too many contaminants in the water bodies. Animals and plants using contaminated water may also be affected. Toxic substances

accumulate with time in the aquifers and once the prime spreads it may render the groundwater unsuitable for human and animal consumption. The effects are serious especially in people who rely on groundwater during drought periods.

Solutions of Groundwater Pollution

1. Legislation

There are federal laws in most countries that help in protecting the quality of groundwater. Safe Drinking and Clean Water regulations should ensure the protection of drinking water by establishing measures for them to meet health standards.

2. The use of water cleaning systems

Point-of-use treatment systems should be installed in outlets that dispense water for human consumption. The techniques used include chemical disinfection, boiling, solar distillation, filtration, ozone water disinfection, activated charcoal absorption, and ultraviolet disinfection.

Arsenic Removal Filters (ARFs) are usually installed to remove arsenic compounds present. Maintenance of these filters is essential to ensure that the drinking water is always safe.

Groundwater Remediation is also another management technique. The biological treatment techniques employed are bio augmentation, bio slurping, bioventing, phytoremediation, and biosparging. Chemicals techniques such as ion exchange, ozone gas injection, membrane separation, and chemical precipitation can also be used.

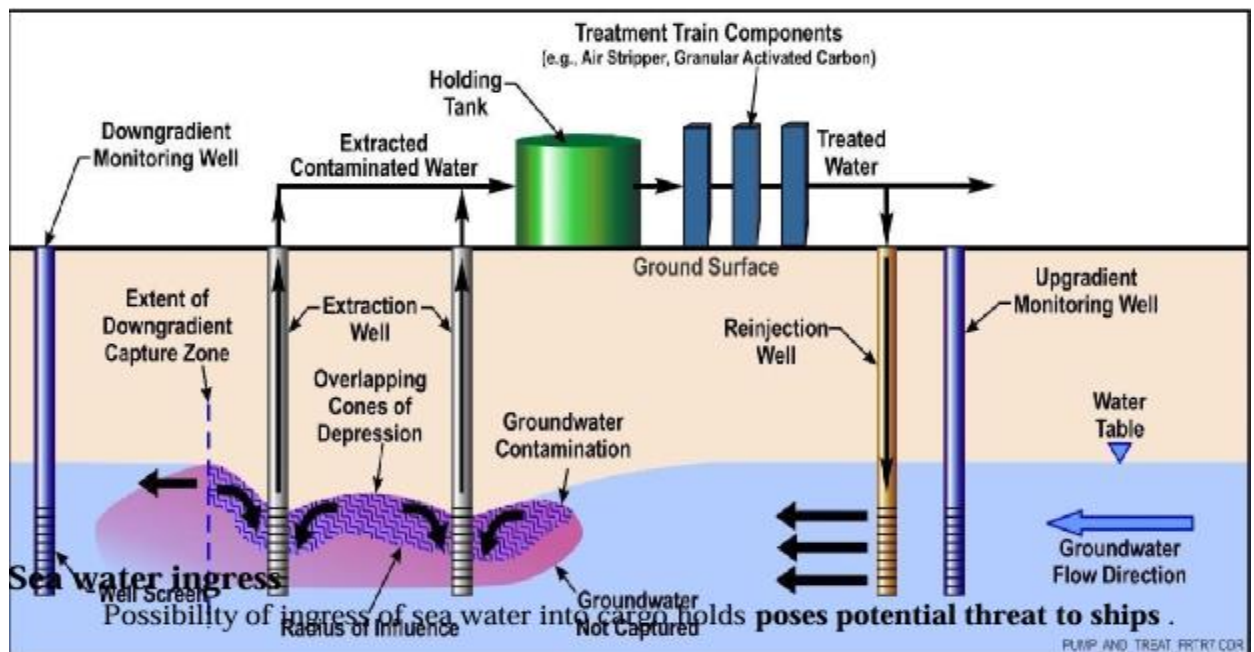
3. Proper management of the sources of pollution

The landfills should be designed with proper clay and leachates. The maintenance should be done regularly. The location of the landfill should also be far from groundwater areas. Further, any hazardous wastes should not be dumped in the landfill unless it is designed for that purpose. In constructing and managing underground storage tanks, it is important to comply with the set regulations and policies to avoid contamination or even lawsuits. A containment device that acts as a leak back up should be put in place and any unused underground tanks should be removed. Underground pipelines installation should be designed professionally. Inspections should be done regularly and causes of corrosion or leakages noted should be resolved immediately.

4. Recycling

Most landfills in various countries have a recycling plant nearby. Therefore, used petroleum products should be taken to such places. Apart from oil, other recyclable materials such as plastic, bottle, and paper wastes can also be taken to recycling plants. The state should provide designated recycling pick up areas in places that they are not established.

Together with other environmental organizations, the state can mobilize people to participate in the recycling initiative. They can do this by holding awareness campaigns and educating communities on the importance of recycling.



Masters of bulk carriers should be aware that the purpose of the Water Ingress Monitor (WIM) is to provide the maximum possible early warning of a condition that may seriously threaten the vessel's survival.

Bulk carriers potential hazards

Ingress of sea water into cargo holds poses significant threat to bulk carriers while on sea passage and in harbour. Effective means of monitoring hold bilges for presence of water are traditional practice. However New regulations now become part of the Safety of Life At Sea (SOLAS) convention and came into force on 1 July 2004. According to this regulation bulk carriers are now required to carry equipment that will give early warning of water ingress to the hull.

Sea Water Ingress Monitoring

Water Ingress Monitoring (WIM) is not a new concept. Masters will recognize that daily monitoring of bilges and tanks has been a feature of prudent seamanship since antiquity.

However, this method of monitoring does not provide continuous information.

Furthermore, when weather deteriorates the manual process is usually suspended because of the dangers to crew members taking the soundings. In such conditions the risk of flooding in the ship is increased and therefore some method of continuous monitoring should be introduced.

The concept now adopted for bulk carriers by (International Maritime Organization) IMO monitors not only for the presence of water, but also, in the cargo hold spaces, the speed of ingress. To achieve this a two stage alarm is used, one at a low level in the hold, the second a short distance above it.

Definition: Sea Water Ingress is the process by which saline water from the sea moves inland into freshwater sources, contaminating groundwater, soil, or surface water and making them unsuitable for drinking, irrigation, or industrial use.

Causes of Sea Water Ingress

Cause	Explanation
1. Over-extraction of Groundwater	Excessive pumping near coast lowers water table, allowing seawater to move in.
2. Rising Sea Levels	Caused by climate change, increases pressure on coastal aquifers.
3. Poor Drainage or Canal Systems	Allows sea water to seep into inland systems during high tide.
4. Land Subsidence	Coastal land sinking due to human activity or natural causes.
5. Cyclones or Storm Surges	Push sea water inland temporarily, causing permanent damage to soil and water sources.
6. Destruction of Natural Barriers	Mangroves, sand dunes, and wetlands act as buffers; their removal increases vulnerability.

Effects of Sea Water Ingress

Area	Impact
Groundwater	Increased salinity makes it unfit for drinking or irrigation.
Agriculture	Soil becomes saline, reducing crop yields and damaging land permanently.
Public Health	Salty water causes health issues like high blood pressure and dehydration.
Infrastructure	Corrosion of pipes, pumps, and buildings due to salt.
Ecosystem	Harms freshwater flora and fauna; loss of biodiversity.

Prevention and Control Measures

Strategy	Description
1. Controlled Groundwater Extraction	Regulate pumping to maintain water table balance.
2. Recharge of Aquifers	Use rainwater harvesting and artificial recharge structures.
3. Mangrove Restoration	Plant and protect mangroves as natural seawater barriers.
4. Construction of Subsurface Barriers	Underground walls or cutoff barriers to block saline intrusion.
5. Use of Salinity-Resistant Crops	In areas where soil is already affected.
6. Monitoring and Zoning	Regular salinity checks and buffer zones near the coast.

Problems	Solutions
Groundwater salinization	Controlled pumping, recharge wells
Soil infertility	Salinity-tolerant crops, soil treatment
Crop damage	Better irrigation management
Health issues	Use treated or alternative water
Corrosion of infrastructure	Use corrosion-resistant materials
Biodiversity loss	Ecosystem restoration, mangroves
Economic losses	Diversified livelihood programs